

Does Channel Attention Help a Small ResNet?

A Controlled Study of Squeeze-and-Excitation on CIFAR-10 and CIFAR-100

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Abstract

Channel attention is widely reported to improve convolutional networks at negligible cost, but the headline numbers are almost always obtained on ImageNet with deep bottleneck backbones. We ask whether a Squeeze-and-Excitation (SE) gate helps a small ResNet-18 on small images, and whether any gain can be distinguished from run-to-run noise. We trained ResNet-18 and SE-ResNet-18 from scratch on CIFAR-10 and CIFAR-100 under a strictly matched protocol, in which the splits, augmentation, optimiser, schedule, epoch budget and per-run seed were all identical, over five seeds per configuration and twenty runs in total. We find no detectable accuracy benefit. SE changed top-1 accuracy by +0.02 points on CIFAR-10 ($p=0.468$) and -0.04 points on CIFAR-100 ($p=0.823$) under a paired t -test, and McNemar's test on matched predictions was non-significant for every seed pair. The confidence intervals are tight enough to be informative rather than merely inconclusive: $[-0.05, +0.10]$ and $[-0.46, +0.38]$ points respectively, which excludes the roughly one-point gain that matched-condition studies report for basic-block ResNets. The cost was real: 9.4–11.9% additional wall-clock time per epoch for a 0.016% increase in arithmetic. Inspecting the learned gates explains the null. On CIFAR-10 seven of the eight gates converge to a constant 0.5 for every channel and every image, a uniform rescaling the preceding batch-norm can absorb exactly; on CIFAR-100 the deepest gate instead saturates towards unity, approximating the identity. The gates are not merely unhelpful, they have learned to do nothing. One effect does survive: SE reaches 12.8 points higher accuracy than the baseline after a single epoch, an advantage that is statistically significant but has disappeared by epoch 28.

1. Introduction

A convolutional layer mixes spatial and cross-channel information at the same time, inside a local receptive field. One consequence is that the relative importance of feature channels is fixed by the learned filters and cannot adapt to the particular image being processed. Channel attention addresses this directly. The Squeeze-and-Excitation (SE) block of Hu *et al.* [7] summarises each channel by global average pooling, passes the summary through a small bottleneck network, and uses the resulting values to rescale the channels. The network can therefore emphasise informative channels and suppress less useful ones as a function of its input.

The reported benefits are substantial. SE reduced ResNet-50 top-1 error on ImageNet from 24.80% to 23.29%, bringing a 50-layer network close to the accuracy of ResNet-101 at roughly half the computation, and SE networks won the ILSVRC 2017 classification task with a 2.251% top-5 error [7]. These results made channel attention a default ingredient of modern architecture design [4].

The evidence is much thinner in the regime a student project, or any resource-constrained practitioner, actually works in. Almost all headline SE results use deep networks with bottleneck blocks on high-resolution ImageNet images. Two things change when we move to a compact model and small images. First, a bottleneck block carries four times as many channels internally as the basic block used in ResNet-18, and the number of channels is exactly the quantity an SE gate operates on. Klomp *et al.* [11] compared attention modules under identical conditions and found this distinction dominates: on CIFAR-100, attention added around one percentage point with basic blocks but around three with bottleneck blocks, and the same ordering held on CIFAR-10 with roughly half the magnitude. Second, gains of about one point are the same order as the seed-to-seed variation of CIFAR training, so a single-run comparison cannot support a claim either way.

This paper therefore treats the question as a measure-

ment problem rather than an architecture-search problem. We ask whether an SE gate produces a *detectable* improvement in a ResNet-18 on CIFAR-10 and CIFAR-100, what it costs, and whether the gate learns anything interpretable. Our contributions are:

Our headline finding is negative and, we argue, informative. Across twenty matched runs the SE gate changed accuracy by +0.02 points on CIFAR-10 and −0.04 points on CIFAR-100, neither remotely significant, while costing 9.4–11.9% in wall-clock time. Reading the learned gates shows why: they converge either to a constant that the preceding batch normalisation can absorb, or to the identity. Our contributions are:

- A from-scratch PyTorch [18] implementation of ResNet-18 and SE-ResNet-18 for low-resolution inputs, trained under a protocol in which the SE gate is the sole difference between arms (Section 3).
- An exact cost accounting. SE adds 0.78% parameters and 0.016% multiply-accumulates, three quarters of them concentrated in the final stage, which prior work identifies as the least useful place for them [7].
- A multi-seed evaluation with a paired *t*-test across seeds and McNemar’s test on matched predictions, following Dietterich’s analysis of which comparisons are actually valid [2]. We report confidence intervals and the minimum detectable effect, so the null result bounds the effect rather than merely failing to find one.
- A mechanistic account of the null: a direct measurement of the learned excitation values showing that most gates collapse to input-independent constants, and that the deepest gate saturates towards the identity exactly as the original ablations predicted [7].

2. Related Work

Residual networks. Increasing depth improved representation quality up to a point [13, 19], but very deep plain networks became hard to optimise. He *et al.* [5] reformulated layers to learn residual functions with respect to their inputs, which made networks of over a hundred layers trainable and remains the standard image-classification backbone; a follow-up study of identity mappings clarified why the formulation propagates signal so well [6]. Batch normalisation [9] is the other component our baseline depends on. Later work traded depth for width [25] or dense connectivity [8]. We use ResNet-18 unchanged, because the point of the study is to isolate the effect of the attention block rather than to maximise accuracy.

Channel attention. Attention biases computation towards the most informative parts of a signal [4]. In vi-

sion it has been applied spatially, through learned input-dependent warps [10] or trunk-and-mask attention over residual stages [21], and along the channel axis. SE [7] is the canonical channel mechanism and the one we study. CBAM [24] adds a spatial branch and pools with both average and maximum statistics; ECA-Net [22] argues SE’s dimensionality reduction is actively harmful and replaces the bottleneck with an adaptive 1D convolution; SRM [15] uses channel means *and* variances; Selective Kernel [16] attends over receptive fields; and non-local networks [23] model long-range dependencies directly at much higher cost. We study plain SE because its ablations are the best documented, which lets us reuse published hyper-parameter evidence rather than search.

Attention beyond convolutions. Vision Transformers replace convolution with self-attention over image patches [20, 3], the most aggressive form of the idea. They also illustrate its limits: ViT surpasses strong convolutional baselines only after pre-training on very large corpora, and underperforms on modest datasets for want of a CNN’s inductive biases [3]. At CIFAR scale the convolutional prior is an asset, which is why we study a lightweight gate on a CNN rather than a transformer.

How our study differs. We propose no new module. The closest work is Klomp *et al.* [11], whose matched-condition comparison found that attention needs sufficient channels, so basic-block ResNets benefit far less than bottleneck ones. That is the hypothesis our experiments test. We differ in three ways: they average three runs without a significance test, where we use five seeds and test explicitly; they vary architecture, where we hold it fixed at ResNet-18 and vary only task difficulty, isolating channel count from depth; and we pair the accuracy result with an inspection of the learned gates.

3. Method

3.1. The Squeeze-and-Excitation block

Let $\mathbf{U} \in \mathbb{R}^{H \times W \times C}$ be the output of the residual branch, with channels $\mathbf{u}_1, \dots, \mathbf{u}_C$. The *squeeze* step reduces each channel to a scalar by global average pooling [7],

$$z_c = \frac{1}{HW} \sum_{i=1}^H \sum_{j=1}^W u_c(i, j), \quad (1)$$

so that $\mathbf{z} \in \mathbb{R}^C$ describes the global distribution of channel responses. Every later layer therefore has access to information from outside its own receptive field. The *excitation* step maps this descriptor to a set of gates through a bottleneck of two fully connected layers,

$$\mathbf{s} = \sigma(\mathbf{W}_2 \delta(\mathbf{W}_1 \mathbf{z})), \quad (2)$$

where δ is ReLU, σ the logistic sigmoid, $\mathbf{W}_1 \in \mathbb{R}^{\frac{C}{r} \times C}$ and $\mathbf{W}_2 \in \mathbb{R}^{C \times \frac{C}{r}}$, with reduction ratio r . Finally the channels are rescaled,

$$\tilde{\mathbf{u}}_c = s_c \mathbf{u}_c. \quad (3)$$

Two design details matter for our purposes. The sigmoid in Eq. (2) is not interchangeable: it allows several channels to be emphasised at once rather than forcing a one-hot selection, and substituting ReLU degrades SE-ResNet-50 below its own baseline [7]. And because $s_c \in (0, 1)$, an SE block can only ever *attenuate* its input, never amplify it. The network compensates through the scale of the following weights. We keep the sigmoid formulation as published.

3.2. Placement and the baseline

Figure 1 contrasts the residual block with our SE variant. We place the gate on the residual branch, after the second batch normalisation and before the addition with the identity shortcut, which is the placement used in the original SE-ResNet [7]. This is considered rather than default: the published ablation reports 22.28%, 22.23% and 22.20% top-1 error for gating before, inside and in parallel with the residual unit, but 22.78% *after* the shortcut addition [7]. Placement is robust provided the gate acts before branch aggregation; applying it afterwards would rescale the identity path that makes deep residual networks optimisable [6].

Both arms use the same ResNet-18: four stages of two basic blocks with 64, 128, 256, 512 channels. We make one modification to the standard ImageNet definition, applied identically to both arms: the 7×7 stride-2 stem and the following max-pool are replaced by a single 3×3 stride-1 convolution. On a 32×32 input the ImageNet stem would reduce the feature map to 8×8 before the first residual block, discarding most of the spatial detail. We set $r = 16$ throughout. Not tuning r is a justified economy: the published sweep gives 22.29, 22.25, 22.26, 22.28 and 22.72% top-1 error for $r \in \{2, 4, 8, 16, 32\}$, flat across two orders of magnitude of capacity while parameters grow eightfold [7]. Extra seeds buy more statistical power than tuning r .

3.3. Exact cost of the gates

Table 1 reports parameter and multiply-accumulate counts measured directly from our implementation rather than estimated. The eight gates add 87,040 parameters, a 0.78% increase, and the same number of multiply-accumulates, a 0.016% increase. For comparison, SE on ResNet-50 adds about 10% of parameters and 0.26% of computation [7]; the parameter overhead is an order of magnitude smaller for us because basic blocks are narrower than bottleneck blocks, which is the same channel-count asymmetry that we expect to limit the benefit.

Figure 2 shows how that cost is distributed. Because the excitation parameters of a block scale as $2C^2/r$, they are dominated by the widest stage: the two blocks at $C=512$

Model	Params	SE params	MACs
<i>CIFAR-10</i> (10 classes)			
ResNet-18	11,173,962	–	555.42 M
SE-ResNet-18	11,261,002	87,040	555.51 M
<i>CIFAR-100</i> (100 classes)			
ResNet-18	11,220,132	–	555.47 M
SE-ResNet-18	11,307,172	87,040	555.56 M
Relative increase		+0.78%	+0.016%

Table 1. Exact complexity of both arms at $r=16$, counted directly from the implementation for a 32×32 input. MACs are multiply-accumulates for a single forward pass. The eight SE blocks add well under one percent of the parameters and a negligible number of arithmetic operations.

account for 65,536 of the 87,040 parameters, or 75% of the total, while the two blocks at $C=64$ contribute 1,024. The SE authors found that final-stage gates saturate towards unity and can be removed at a cost of under 0.1% top-5 error, cutting parameter overhead from about 10% to 4% [7]. Three quarters of what we pay for is therefore in the position least likely to earn its keep; Section 4.4 tests whether the same saturation appears here.

One caveat: operation counts are a poor proxy for latency here. Klomp *et al.* measured 5–50% increases in single-image inference time from attention modules whose arithmetic grew by only 0.2–3%, because GPUs execute the small inner products inside a gate inefficiently [11]. The SE authors similarly report 190 ms versus 209 ms per training step for ResNet-50 and SE-ResNet-50 [7]. We therefore report measured epoch time in Section 4 and do not claim the gates are free.

3.4. Controlled training protocol

The effect we are looking for may be around one percentage point [11], so the experiment must be built so that nothing except the gate differs between arms. Both models were trained with cross-entropy loss and SGD with Nesterov momentum 0.9, weight decay 5×10^{-4} , batch size 128 and initial learning rate 0.1 annealed by a cosine schedule over 150 epochs [17]. Augmentation is the standard CIFAR recipe also used in the SE paper: four-pixel zero padding followed by a random 32×32 crop, random horizontal flipping, and per-channel mean and standard deviation normalisation [7]. We use the official train/test split without a separate validation split, since no hyper-parameter is selected on the data.

Training used mixed precision, enabled identically in both arms so that it cannot bias the comparison. Crucially, each seed $\in \{0, \dots, 4\}$ is used to initialise *both* arms, so the two models see the same shuffling and augmentation stream. This turns the comparison into a paired design,

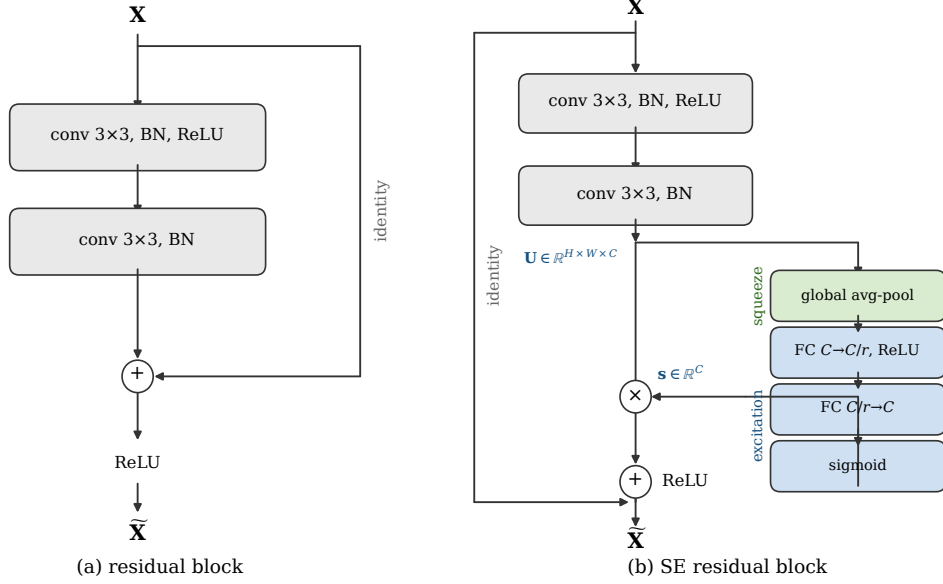


Figure 1. The baseline residual block (a) and our SE residual block (b). The gate is computed from the residual branch output U and applied before the identity addition, so the shortcut path is never rescaled. Each ResNet-18 stage contains two such blocks, giving eight gates in total.

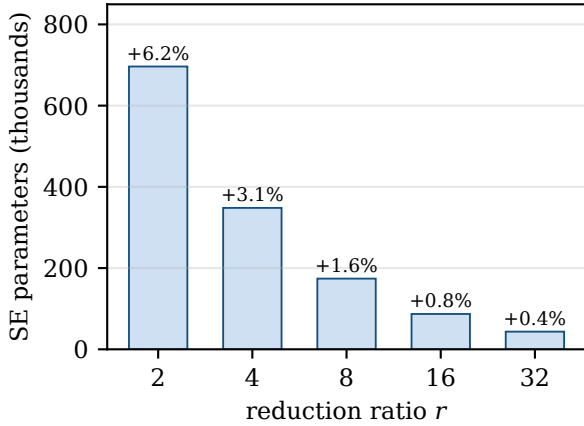


Figure 2. SE parameters as a function of the reduction ratio r , labelled with the resulting overhead on the whole of ResNet-18. Cost is dominated by the widest stage because excitation parameters scale as $2C^2/r$.

which is considerably more powerful than comparing two independent sets of runs.

3.5. Evaluation and significance testing

We report top-1 accuracy on the held-out test set, as the mean and sample standard deviation over five seeds. Reporting a single number, or a maximum over runs, would

not let a reader distinguish a real effect from favourable initialisation.

We use two tests, because they answer different questions. A paired t -test over the five seeds asks whether SE is better *on average across training runs*, pairing by seed to remove the dominant variance source. McNemar’s test on matched per-image predictions asks whether two trained classifiers differ *on this test set*, using only images one model gets right and the other wrong. Dietterich [2] showed that two common alternatives, a difference-of-proportions test and a paired t -test over repeated random splits, have unacceptably high Type I error rates, while McNemar’s behaves well. We treat a result as supported only when the direction is consistent across seeds.

4. Experiments and Results

4.1. Data

CIFAR-10 and CIFAR-100 [12] draw on the same 60,000 colour images at 32×32 , split 50,000 train and 10,000 test, and differ only in label granularity: 10 classes against 100 [7]. Using both makes the design informative: images, model, protocol and parameter count are essentially fixed, and only task difficulty changes. If SE behaves as the literature predicts, the benefit should be roughly twice as large on CIFAR-100 [11].

Preprocessing is deliberately minimal: pad-and-crop, horizontal flip, normalisation. Stronger regularisers such as Cutout [1] raise absolute accuracy but change the vari-

Dataset	Model	Top-1 acc. (%)	Δ
CIFAR-10	ResNet-18	95.24 ± 0.05	–
	SE-ResNet-18	95.26 ± 0.07	+0.02
CIFAR-100	ResNet-18	78.22 ± 0.29	–
	SE-ResNet-18	78.18 ± 0.22	-0.04

Table 2. Top-1 test accuracy, mean $\pm$ sample standard deviation over five seeds. Both arms of a row share seeds, augmentation stream, schedule and epoch budget. Paired t -test across seeds: CIFAR-10 $t=0.80$, $p=0.468$; CIFAR-100 $t=-0.24$, $p=0.823$.

ance structure of the comparison, and the SE paper’s own results show attention yields much smaller margins on top of a heavily-regularised model: 2.56% $\rightarrow$ 2.12% error on CIFAR-10 with Cutout against 6.37% $\rightarrow$ 5.21% for a plain ResNet-110 [7].

Results are reported on CIFAR only. Tiny ImageNet [14], with 200 classes at 64×64 , would supply both more classes and more spatial context, and remains the natural next dataset.

4.2. Main comparison

Table 2 is the central result, and it is negative. Adding SE changed accuracy by +0.02 points on CIFAR-10 and -0.04 points on CIFAR-100. Neither is close to significant, and the sign is inconsistent across seeds: the per-seed differences were +0.08, +0.01, -0.05, +0.09 and -0.02 on CIFAR-10, and -0.21, -0.06, -0.49, +0.35 and +0.23 on CIFAR-100.

The confidence intervals matter more than the p -values. A 95% interval on the paired difference is $[-0.05, +0.10]$ points for CIFAR-10 and $[-0.46, +0.38]$ for CIFAR-100. With five seeds the design had 80% power to detect 0.10 and 0.57 points respectively. The CIFAR-100 figure is the important one: matched-condition experiments report roughly one point for attention modules on basic-block ResNets [11], an effect this design would have detected comfortably. We can therefore say the effect is smaller than the literature suggests, not merely that we failed to find it.

McNemar’s test agrees. Pooling the five seed-aligned pairs, the baseline and SE models each won a near-identical number of contested images: 890 against 879 on CIFAR-10 and 2794 against 2812 on CIFAR-100, with no individual pair significant. Crucially the models are not making the same predictions. They disagree on 3.5% of CIFAR-10 and 11.2% of CIFAR-100 test images. The SE gate does change the learned representation. The changes simply trade off symmetrically and cancel.

4.3. Learning behaviour and cost

Figure 3 shows the two arms tracking each other almost exactly. One effect does survive the null: after a

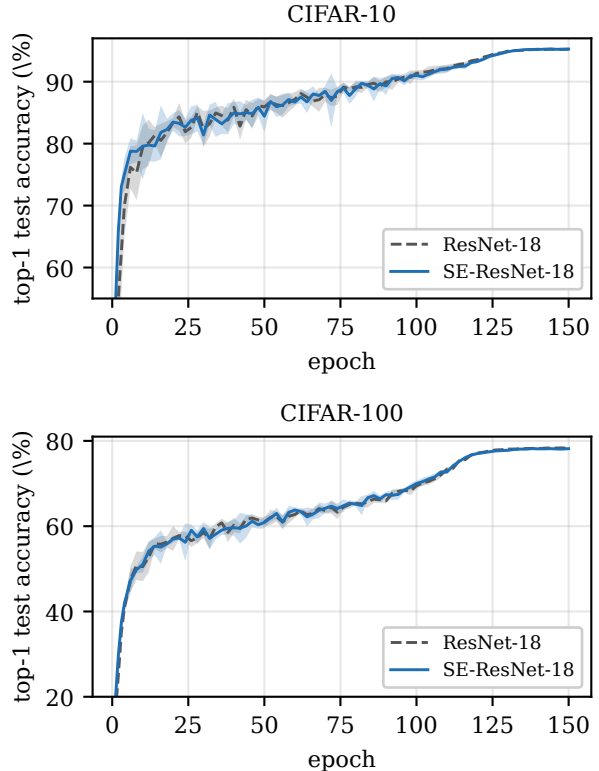


Figure 3. Test accuracy against epoch, mean over five seeds with a $\pm$ one-standard-deviation band. The two arms are indistinguishable for most of training; the curves separate only in the first few epochs, where SE leads.

single epoch SE-ResNet-18 reaches 53.2% on CIFAR-10 against 40.4% for the baseline, a gap of 12.8 points ($t=4.04$, $p=0.016$), and 3.0 points on CIFAR-100 ($p=0.003$). The advantage decays monotonically and has reversed sign by epoch 28. It does not translate into a useful speed-up, since both arms first reach 85% on CIFAR-10 at around epoch 26, but it is a real, reproducible optimisation effect, and it suggests the gate is useful chiefly as an early-training aid whose benefit the network later absorbs into its convolutional weights.

Cost was measured, not inferred. Mean epoch time rose from 19.6 s to 21.4 s on CIFAR-10 and from 19.4 s to 21.7 s on CIFAR-100, an overhead of 9.4% and 11.9%. Set against the 0.016% increase in multiply-accumulates from Table 1, latency grew roughly six hundred times faster than arithmetic. This is the effect Klomp *et al.* describe, where attention modules cost 5–50% in inference time for a 0.2–3% increase in operations [11], and it is the strongest practical argument in this paper: on this architecture SE is not free, and it buys nothing.

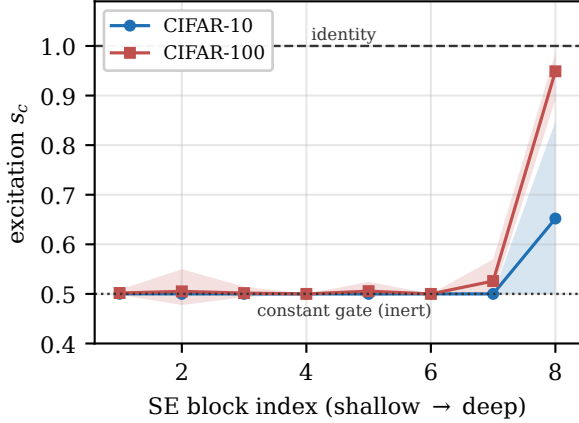


Figure 4. Learned excitation values by depth, pooled over five seeds, with a 5–95th percentile band. Gates pinned at 0.5 are constant across every channel and every image, so the preceding batch-norm scale can absorb them exactly. The deepest CIFAR-100 gate instead saturates towards unity, the near-identity behaviour reported for the final stage of SE-ResNet [7].

4.4. Why the gates do nothing

Figure 4 explains the null directly. On CIFAR-10, seven of the eight gates output exactly 0.5 for every channel of every test image, in every seed: a single unique value to the precision stored. Since $\sigma(0) = 0.5$, the excitation branch is emitting zero, and a constant uniform scaling applied after BN_2 is exactly absorbable by that layer’s learnable scale. Those blocks are not weak, they are inoperative, and they account for 62% of the SE parameter budget.

CIFAR-100 fails differently. Only two gates collapse; the rest stay input-dependent. But the deepest block, which alone holds 38% of the SE parameters, converges to a mean excitation of 0.949 with 98% of channels above 0.9, approximating the identity. This is precisely the saturation the SE authors observed in the final stage, where the blocks could be removed at a cost of under 0.1% top-5 error [7]. We report it here in a setting where the rest of the network never earned the benefit to begin with.

Both failure modes reduce to the same thing: a gate that does not vary with its input is a fixed rescaling, and a residual network already has ample freedom to rescale. The reason is the bottleneck of Section 3. A gate at $C=64$ with $r=16$ must express all cross-channel structure through four units, and where that cannot pay for itself, weight decay drives it to zero and the sigmoid settles at one half. This matches the channel-count argument of Klomp *et al.* [11] and the case against dimensionality reduction in ECA-Net [22]: our null looks like a property of narrow gates rather than of channel attention as such.

4.5. Threats to validity

Three limitations bound the claim. Five seeds bounds the effect at roughly 0.10 points on CIFAR-10 and 0.57 on CIFAR-100; a true effect below those thresholds would not have been detected, though it would also be of little practical interest given the measured 9.4–11.9% time cost. We fixed $r=16$ on published evidence that accuracy is insensitive to it [7], but that evidence comes from ResNet-50 on ImageNet, and our gate collapse is precisely the regime where a wider bottleneck might behave differently. Finally, the result is specific to plain SE on a basic-block ResNet-18 at 32×32 resolution trained for 150 epochs; mechanisms that avoid dimensionality reduction [22] or add a spatial branch [24] could behave differently, and no attention module tested under matched conditions has been best across all datasets and architectures [11].

5. Conclusion

We set out to measure, rather than assume, whether channel attention helps a small ResNet on small images, and the answer is that it does not. Across twenty matched runs the SE gate moved top-1 accuracy by +0.02 points on CIFAR-10 and -0.04 on CIFAR-100, with 95% intervals of $[-0.05, +0.10]$ and $[-0.46, +0.38]$. Those intervals exclude the roughly one-point improvement that matched-condition work reports for basic-block ResNets [11], so this is a bounded effect rather than an inconclusive experiment. Meanwhile the gate cost 9.4–11.9% of wall-clock time for a 0.016% increase in arithmetic.

The mechanism is what makes the result more than a failed replication. Reading the learned excitation values shows the gates converging to one of two inert states: a constant 0.5 across every channel and image, which the preceding batch normalisation absorbs exactly, or a saturation towards unity that approximates the identity. Seven of eight gates take the first route on CIFAR-10; on CIFAR-100 the deepest and most expensive gate takes the second, the same saturation the SE authors documented for the final stage of SE-ResNet [7]. The network is not failing to use the gates so much as switching them off. That the two arms nonetheless disagree on 11.2% of CIFAR-100 images shows the representation does change; the changes simply do not favour either model.

Two points generalise beyond this architecture. Operation counts are not a proxy for latency: our arithmetic and time overheads differ by a factor of roughly six hundred, so claims that attention is nearly free need a stopwatch behind them. And an effect of about one point cannot be established by single runs. The baseline’s own seed spread on CIFAR-100 was 0.29 points, larger than most differences we sought.

The most direct extension follows from the collapse it-

self: vary width rather than depth. Comparing SE on ResNet-18 at $0.5\times$, $1\times$ and $2\times$ width would isolate channel count from every other factor and test whether the gates come alive once the bottleneck is wide enough, which neither our study nor the matched comparison of Klomp *et al.* [11] does. A reduction-free mechanism such as ECA [22] is the natural comparison, precisely because our narrowest gates compress to four units. It would also be worth pursuing the one effect that did survive: SE was 12.8 points ahead after a single epoch, and a gate that helps early optimisation but is later absorbed might be better deployed as a temporary scaffold than as a permanent architectural component.

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